



FactNews



Natural Language Processing - Assignment 2

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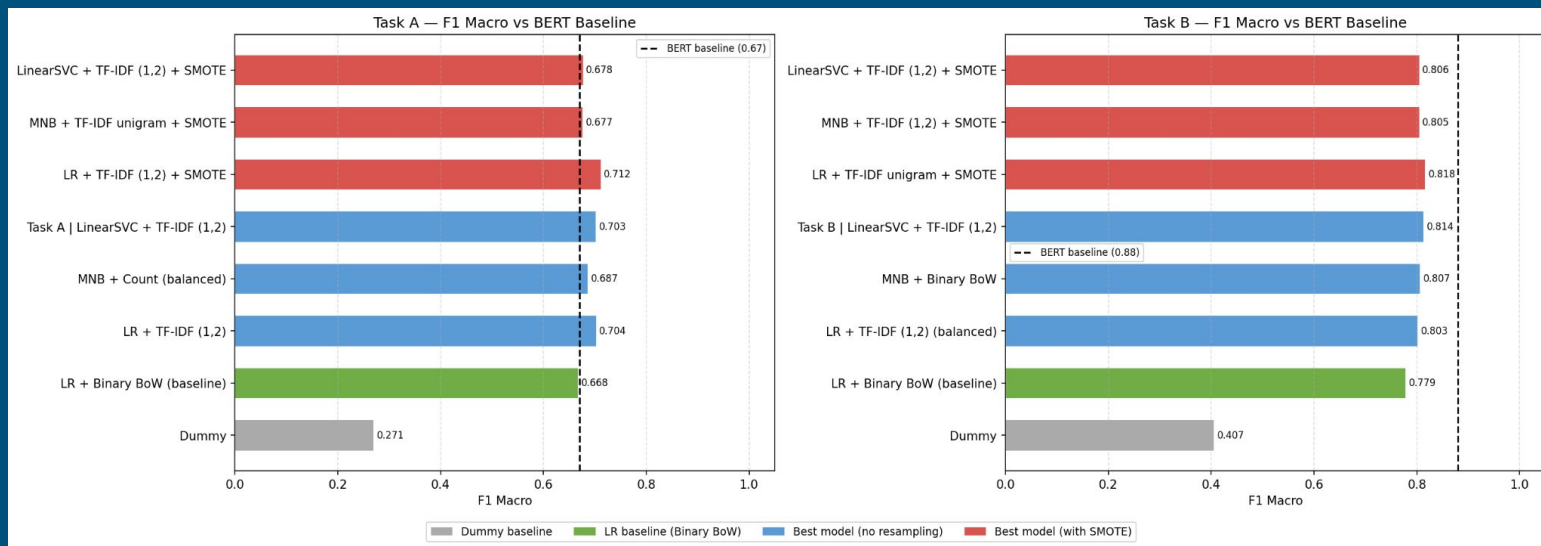
Data Provenance and Characteristics

Dataset from the **FactNews** repository and the accompanying RANLP 2023 paper:

- In Brazilian Portuguese, contains 6,191 annotated sentences, and comes from 300 news documents covering 100 news stories
- Uses three media outlets: **Folha de São Paulo**, **Estado**, and **O Globo**
- Covers domains such as politics, world, sports, daily, culture, and science
- Includes sentence-level annotations for:
 - **factual** as 0
 - **biased** as 1
 - **quotes** as -1

Assignment 1 Results

Fig: Assignment 1 results



The LR + TF-IDF (1,2) + SMOTE is the best model with 0.712

Models - mBERT & BERTimbau

mBERT

bert-base-multilingual-cased

Architecture	BERT-base
Parameters	~110 M
Pre-training	104 languages, Wikipedia
Portuguese	Both PT-BR and PT-PT
Purpose	Multilingual baseline

Fine-tuning approach

- Standard seq. classification head (3-class)
- Class-weighted cross-entropy loss
- $\text{lr} = 2\text{e-}5 \cdot \text{batch} = 16 \cdot \text{epochs} = 3$

BERTimbau

neuralmind/bert-base-portuguese-cased

Architecture	BERT-base
Parameters	~110 M
Pre-training	PT-BR - BrWaC + Wikipedia
Portuguese	Brazilian variant only
Purpose	Strong PT-BR baseline

Fine-tuning approach

- Same head as mBERT + class weights + Resampling variants
- $\text{lr} = 2\text{e-}5 \cdot \text{batch} = 16 \cdot \text{epochs} = 3$

BERTimbau + MLM + LoRA + CW

Our best performing model combined the strongest Portuguese pre-trained backbone with a three-step enhancement pipeline:

- **1. Domain Adaptation (MLM):** Continued pre-training on unlabelled Brazilian news (including LLM-generated texts) to capture domain-specific vocabulary and journalistic style.
- **2. LoRA (Low-Rank Adaptation):** Parameter-efficient fine-tuning that reduces the number of trainable parameters, acting as a strong implicit regularizer. By significantly mitigating the risk of overfitting, LoRA allowed us to safely train for more epoches, ultimately achieving better results while preserving the model's deep pre-trained knowledge and preventing catastrophic forgetting.
- **3. Class Weighting (CW):** Modified the loss function to heavily penalize errors on minority classes (e.g., "biased" news), naturally handling the dataset's severe imbalance.

Source	Description	Size (Sentences/Texts)
FactNews (<code>train_df</code> + <code>test_df</code>)	The labelled dataset used throughout this notebook	6,191
<code>df_domain_adap</code>	An additional unlabelled Brazilian news corpus (<code>conteudo_noticia</code> column)	10,081
<code>df_domain_adap_two</code>	A second unlabelled news corpus from Folha de São Paulo (<code>text</code> column)	165,902

Models - Albertina PT-BR & Zero-Shot

Albertina PT-BR (Recommended)

PORTULAN/albertina-900m-portuguese-ptbr-encoder-brwac

Architecture	DeBERTa-v2-XL
Parameters	~900 M
Pre-training data	BrWaC + PT-BR web (PORTULAN)
Language	Brazilian Portuguese
Tokenizer	SentencePiece, PT-BR vocab

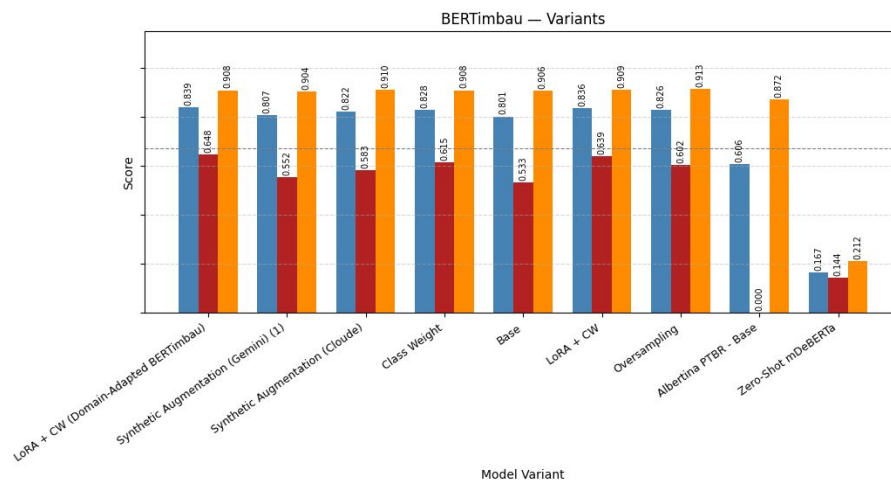
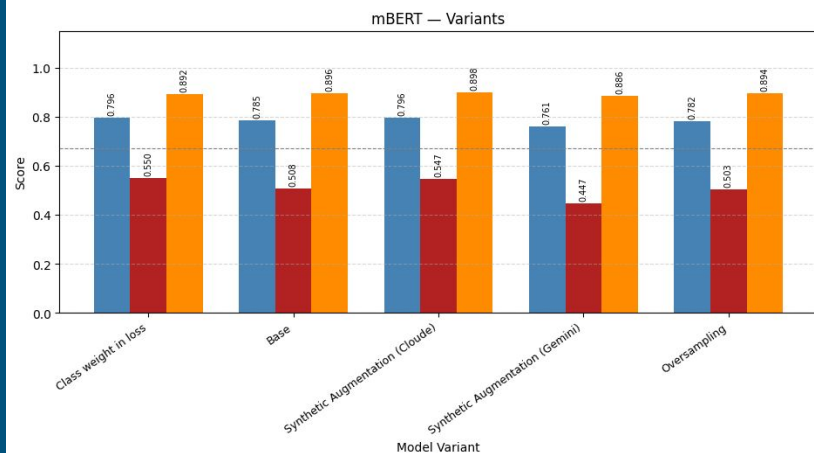
Zero-Shot

MoritzLaurer/mDeBERTa-v3-base-mnli-xnli

Architecture	DeBERTa-v3-base
Pre-training data	CC100 multilingual dataset. by Microsoft
Language	100 (pre-training)
Task	Natural Language Inference

Experimental Results

Task A — Model Comparison by Base Architecture (mBERT vs BERTimbau)



----- BERT baseline (F1 Macro=0.67) F1 Macro F1 Biased Accuracy

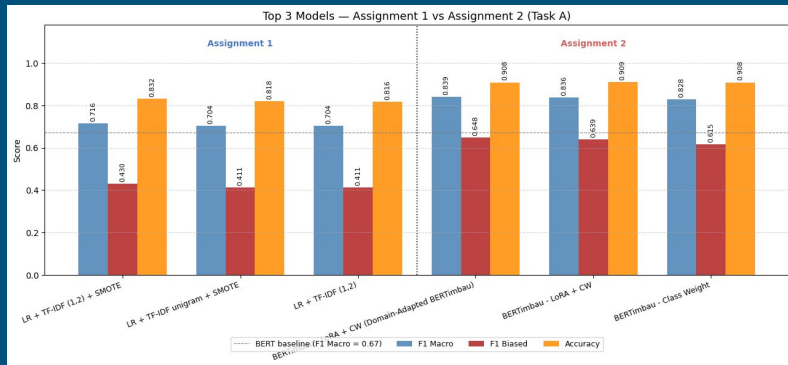
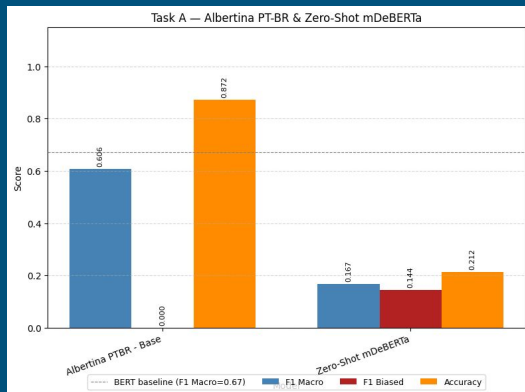
BEST MODEL

0.8386

BERTimbau LoRA+CW (DA) - F1 Macro

↑ All transformers outperform the Assignment 1 best (0.71).

Experimental Results



BEST MODEL

0.8386

BERTimbau LoRA+CW (DA)

vs Assignment 1

+0.21

Macro-F1 improvement



Albertina PT-BR

biased F1 = 0.000

Model predicts biased=0 for all test samples.

Per-class pattern (all models)

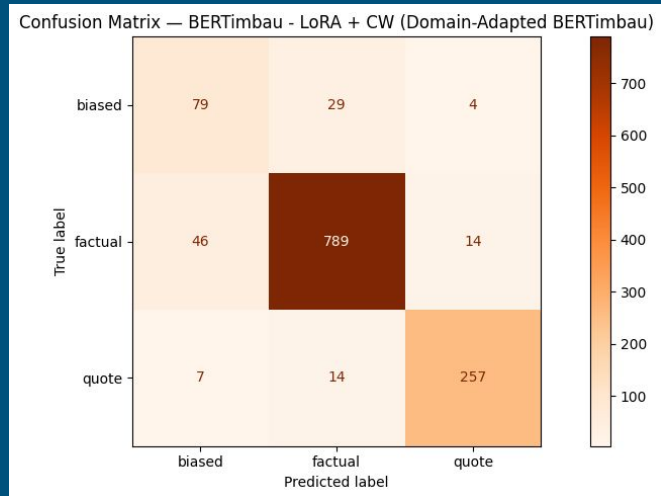
factual: 0.92–0.94

quote: 0.90–0.94

biased: 0.00–0.65

Error Analysis

In every model, the majority of errors occur in the *biased* row, where a significant fraction is misclassified as *factual*. This is linguistically expected, biased sentences share surface structure with factual ones, making the *biased/factual* boundary ambiguous. By contrast, *factual* and *quote* are well separated from each other, with very few cross-errors between them.



#	Bias Type	Description
1	Spin	Interpretation that goes beyond the facts reported
2	Word Choice	Use of emotionally charged or loaded vocabulary
3	Subjective Adjectives	Adjectives that reflect opinion rather than fact
4	Mind Reading	Attribution of thoughts or intentions without evidence
5	Generalization	Broad claims drawn from insufficient evidence
6	Omission	Selective exclusion of relevant information
7	Placement	Strategic positioning of information to influence perception
8	Story Selection	Choice of which stories to cover or ignore
9	Source Selection	Choice of which sources to quote or omit
10	Labeling	Use of ideologically charged labels for people or groups
11	Framing	Presenting facts within a particular ideological context
12	Sensationalism	Exaggeration or dramatization of events

Conclusion

01

Transformers outperform traditional ML

All fine-tuned transformers surpass the best Assignment 1 models. Contextual representations capture nuances that TF-IDF cannot.

02

Albertina PT-BR performance

Underperformed due to hardware constraints, not model weakness, causing complete class collapse on "biased". Full-precision training on adequate hardware would be expected to yield better results.

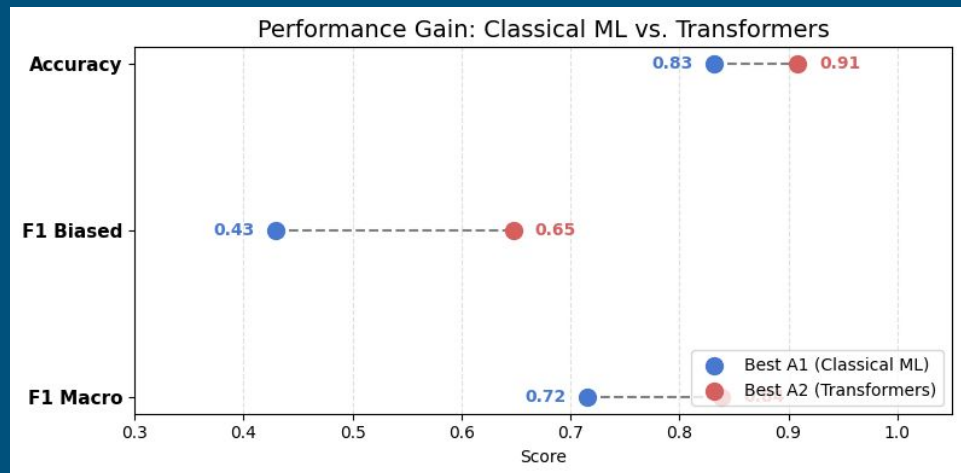
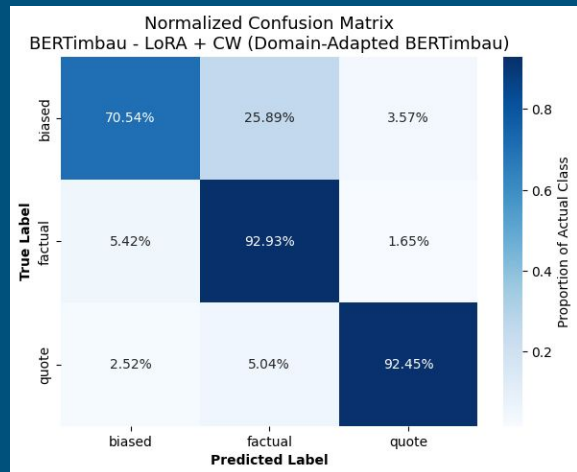
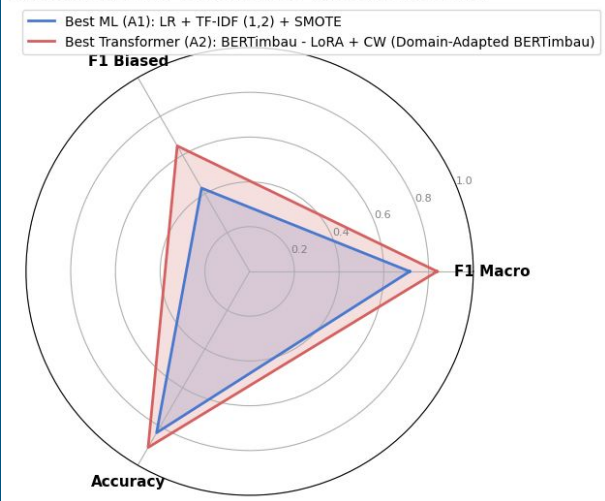
03

Bias detection remains hard

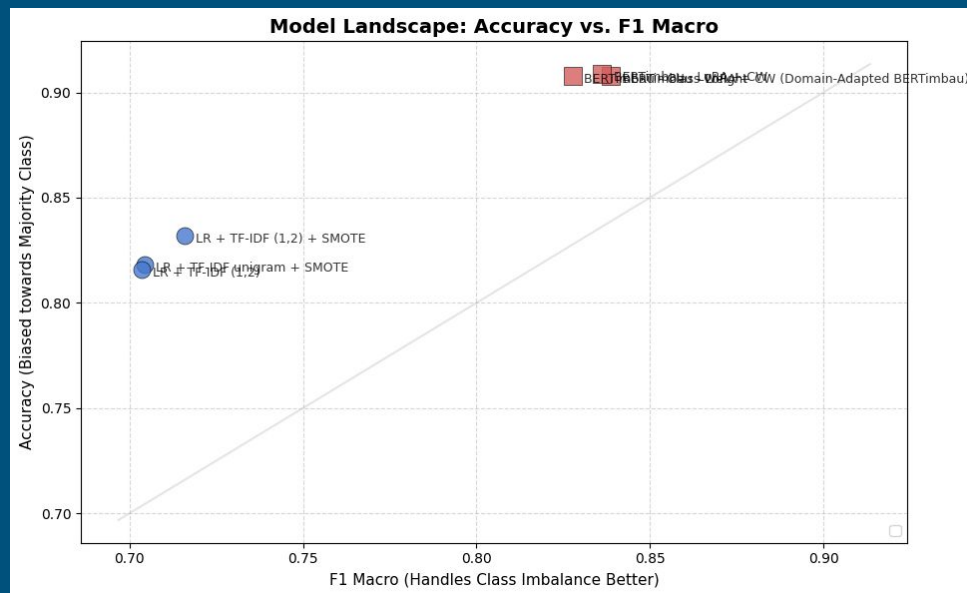
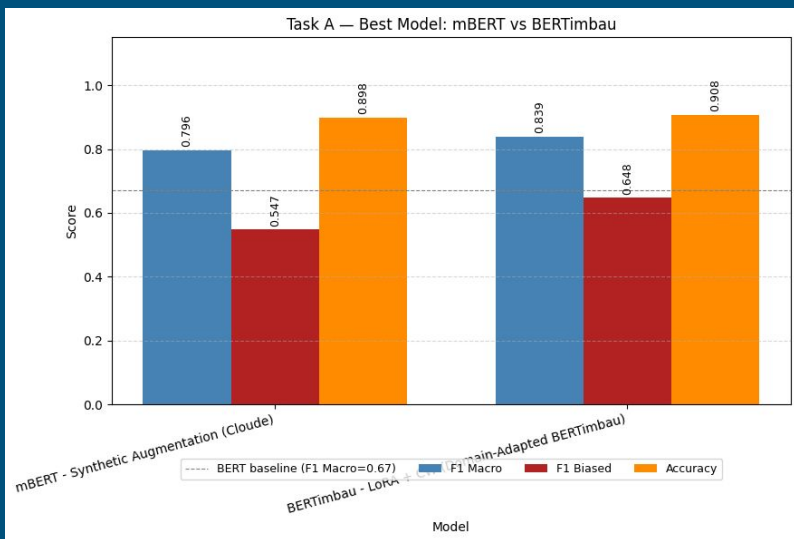
The 'biased' class is systematically the weakest across all models. More annotated data and targeted augmentation are needed.

Annexes

Multidimensional Performance: ML vs. Transformer

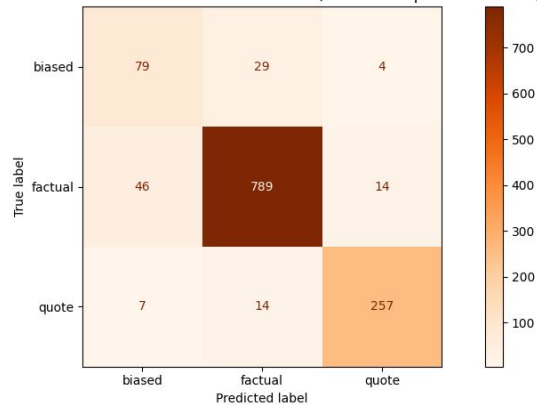


Annexes

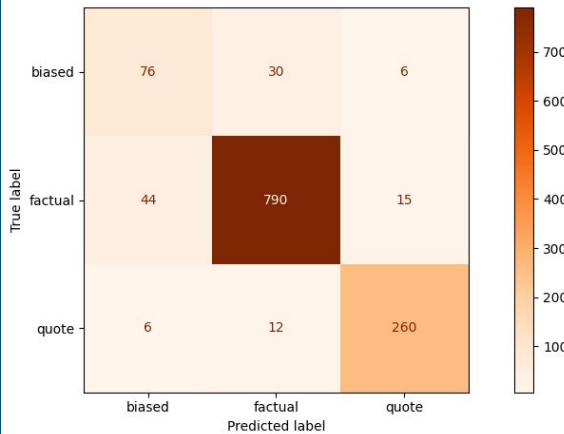


Annexes

Confusion Matrix — BERTimbau - LoRA + CW (Domain-Adapted BERTimbau)



Confusion Matrix — BERTimbau - LoRA + CW



Confusion Matrix — mBERT - Synthetic Augmentation (Cloude)

